



ISRO sets its sights on Venus

ISRO is planning to send an orbiter to Venus in 2023 and is currently looking for international collaborators. <https://digit.in/dec18-70-1>



A cloud looms over Mars

A 1500-km long cloud looms over the Arsia Mons volcano on Mars, with staying power unlike other clouds. <https://digit.in/dec18-70-2>

TOUCHING THE SUN

Three months after its launch, the Parker Solar Probe became the fastest man made object in the solar system

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n 29 October, the Parker Solar Probe reached within 42.6 million kilometres of the Sun, the closest

any man made object has been to the engine that powers the Solar System. Just a few days later, on 6 November, the spacecraft made its closest approach to the Sun, reaching a point in its current orbit known as the

perihelion. The instrument was moving at a speed of 343,000 kilometres per hour, smashing the record for the fastest spacecraft. The previous record was held by the Juno spacecraft, on its ultimate dive into Jupiter, at 266,000 kilometres per hour. The Parker Solar Probe is just getting started though. Over a series of orbits, it will become faster and faster, diving deeper into the corona in each circuit. At its first close approach, the probe was at a distance of 24 million kilometres from the Sun. Because of the intense radio emissions from the Sun, the spacecraft remained out of contact from the Earth for a few days. Advanced autonomous systems on board kept the spacecraft up and running, and safe from the sun, during the time it was out of contact with home.

THE MYSTERIES OF THE SOLAR CORONA

The corona is the part of the Sun that is still visible during a Solar eclipse. The high energy particles of the corona are so hot, that the temperatures are higher than even the surface of the Sun. The spacecraft is an attempt to learn more about the origin of much of the space weather in the solar system, by diving into the sun. The spacecraft will be exposed to temperatures in excess of one million degrees Celsius. Protecting the sensitive instruments on board, is a 4.5 inch thick, and 2.4 metre wide heat shield. The shield is made up of a carbon composite foam sandwiched between two carbon plates. The Sun facing side of the shield has a coating of white ceramic paint, to reflect away as much sunlight as possible. The instruments on board will remain at a relatively cool 30 degrees Celsius, thanks to the shield.

There are some instruments that do peek out around the shield though, such as antennas and a particle collecting Faraday cup. These are necessary to make the observations. Also